

CONCEPT CASE STUDY — NOT AN OFFICIAL SPOTIFY PRODUCT

Listening Rhythm

A UX/Product Case Study: Listener Burnout Detection for Spotify

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1. Executive Summary

No or maybe one or two published case studies currently document Spotify detecting **listener burnout** specifically. The closest real research happens to be — most notably the University of Bristol's *Mood Music* study, which linked Spotify streaming history to depression, anxiety like mental health issues and well-being scores across 171 students — but nothing brings out this as an in-app feature.

This case study is supposed to fill that gap. It proposes **Listening Rhythm**, a passive and opt-in well-being signal built entirely from listening *behavior* (not lyrics analysis, not self-report or biometric data) that surfaces gentle, non-diagnostic nudges when a user's patterns resemble burnout: late-night binge sessions, rising skip rates, narrowing genre diversity, and looping the same tracks that might be low in energy or fall under the slow and sad genre.

The design is built around one - single hard constraint: **a music app should never play therapist, it should be just there to make a user feel light-hearted.** Every decision below — copy, visual language, escalation path — is shaped by that constraint.

2. Problem Framing

Why this belongs on a music platform

People already use music as an unconscious mood regulator. Research on stress recovery has shown that self-selected music listening after stressful events groups into different patterns which include calmer, sadder, more minor-key tracks that differ measurably from baseline listening. Spotify already has this signal for every user but beyond providing recommendations, it does not do anything.

Why “burnout” specifically

Burnout is behavioral before it is clinical. It shows up as:

- **Escape-listening:** long, late-night sessions with minimal engagement (no skips, no searches, just letting it run)
- **Numbing loops:** repeating the same 3-5 tracks for days
- **Energy collapse:** a sustained drop in tempo/energy/valence of chosen tracks
- **Fragmented attention:** rising skip rate, shorter session length, more genre-switching

None of these might *prove* burnout. But as a **pattern**, they prove to be a legitimate, non-invasive signal which is closer to how a smartwatch flags irregular sleep than to a diagnosis.

The trap to avoid

The obvious failure mode is a feature that strikes “You seem depressed” presumptuous, clinical, creepy and panicking at platform scale. The design must earn trust by staying **descriptive instead of diagnostic**, reflecting listening patterns back to the user and never naming a mental state for them.

3. Research Grounding

Source	Relevance
<i>Mood Music</i> (Univ. of Bristol; EMA + Spotify history + PHQ-9/GAD-7/SWEMWBS, $n = 171$)	Confirms streaming metadata correlates with real-time mood; supports using listening data as a <i>proxy signal</i> , not a diagnostic tool
Stress-recovery listening studies	Shows audio-feature shifts (energy, valence, tempo) are a measurable, real phenomenon post-stress - the technical basis for a “rhythm” signal
Public playlist analysis (search terms like “burnout,” “overthinking,” “de-stress”)	Confirms users already self-label listening by emotional state, suggesting language like “decompress” and “reset” will feel native rather than clinical

Design implication: The feature should read signals people are already using implicitly (they *already* build “de-stress” playlists) rather than inventing or giving it a new clinical term.

4. Persona

Priya, 29, product designer. A Premium user, with listening to music for 3+ hours in a day. For the last few weeks she has been going to bed late, her “Discover Weekly” engagement has dropped, and she has been replaying the same lo-fi playlist every night instead of skipping through it like she used to earlier. She hasn’t consciously noticed any of this. She would be irritated by a pop-up asking “Are you okay?” but might appreciate a quiet, optional insight that are on her own terms and she can dismiss it in one tap.

Priya is the target: **not someone in crisis**, but someone in a slow behavioral drift who has no reason to notice it herself or is somehow knowingly avoiding to notice this shift in her behavioral pattern.

5. Ethical Guardrails

(non-negotiable constraints on the design)

- Opt-in only**, off by default. No passive shipping of a “we noticed you seem stressed” feature to 600M+ users without consent as this might create a panic among the users.
- Never name a clinical state.** No “depression,” “burnout,” or “anxiety” labels attached to the user. Only describe *behavior* (“your late-night sessions are up”) and let the user draw their own conclusion. This might also help them to realise what they have been doing since past few days.
- No push notifications for this.** It lives in-app, checked only when the user opens the app,

never an interruption.

4. **One-tap exit at every step.** Dismiss, mute for 30 days, or turn off entirely — always visible, never hidden in settings or inside multiple three-dot options.
5. **A support resource is offered, never forced.** If a user chooses to dig deeper, options like grounding exercises or if they ask, professional resources are one tap away but never inserted automatically or repeatedly. Let the users be their own guardians.
6. **No data leaves the feature.** Not shared with advertisers, not used to target ads, stated explicitly in the opt-in screen, since trust is the entire foundation for this feature.

6. The Feature: “Listening Rhythm”

A single visual metaphor carries the whole feature: **your listening as a rhythm/pulse**. Steady, evenly-spaced pulses = a stable pattern. Clustered, late, erratic pulses = a shifting one. It is the same metaphor on every screen, so the feature never needs to explain itself twice.

Signals tracked

(all passive, all already-collected metadata)

- Session start times (late-night clustering)
- Skip rate trend (7-day rolling average vs. personal baseline)
- Repeat-track ratio (same track/playlist looped vs. personal baseline)
- Audio-feature drift (average energy/valence/tempo of chosen tracks vs. personal baseline)

Everything is **relative to the user’s own baseline** and never compared to other users. This avoids both the creepiness of population-level profiling and the inaccuracy of one-size-fits-all thresholds.

User flow

Discover tab (existing) → Opt-in card, shown once “Not now” → never shown again unless user seeks it out “Try it” → passive tracking begins (invisible, no UI change) → Pattern shift detected over rolling window → Quiet insight card appears at bottom of Home feed (not a notification) “Not now” → dismissed, resurfaces no sooner than +14 days “Tell me more” → Weekly Rhythm report (ignored) → disappears after this session, no nagging → Weekly Rhythm report (opt-in deep dive) “Reset playlist” (mood-lift suggestions) “Just data, thanks” (closes, no action) “I’d like support resources” → optional, user-initiated only

7. Key Screens

See the accompanying interactive mockup for the four core screens:

1. **Opt-in card:** plain-language consent, states exactly what is and is not tracked
2. **Home feed insight card:** one line, one visual, two buttons, easily ignorable
3. **Weekly Rhythm report:** the full behavioral picture, framed as data, not diagnosis
4. **Reset playlist:** the one actionable output, turning every insight into something useful immediately

8. Success Metrics

Metric	What it tells us
Opt-in rate	Whether the concept is trusted enough to try
Dismiss-without-return rate	Whether the insight card feels like nagging
“Reset playlist” save rate	Whether the feature offers real value, not just surveillance
Opt-out rate after first insight	The clearest signal of whether tone/trust failed
Session length <i>after</i> engaging with a Reset playlist	Whether the feature actually helps, however loosely defined

We deliberately exclude clinical detection “accuracy” as a success metric. The feature is intended to help users reflect on their listening patterns and overall well-being, not to provide medical assessments or burnout diagnoses.

9. Risks & Open Questions

- **False positives at scale:** A user who has simply started a new night-shift job will trigger the same signals as someone withdrawing. The copy must stay behavioral (“your listening times have shifted”) rather than interpretive (“you seem burnt out”) so a wrong guess does not feel invasive.
- **Regulatory exposure:** Even non-clinical well-being inference from behavioral data increasingly draws scrutiny (EU AI Act “emotion inference” provisions for one). This would need real legal review before ship.
- **Trust is the whole product.** If this ever feels like it’s profiling users for ad targeting, even by rumor; the feature dies overnight. The “no data leaves this feature” guarantee has to be real and independently auditable, not just a line.

10. Why This Belongs in a Portfolio

This case deliberately treats the hardest part of the problem - ethics as the core design challenge, rather than an afterthought. A simpler approach might be to recommend mood-boosting playlist whenever a user feels sad or emotional. But, the real challenge here is to design a feature that genuinely supports users instead of making Spotify feel like it is making assumptions about someone’s mental health. Handling this balance will make this a meaningful product.